NETFLIX DATA ANALYSIS

Project README | Python • Pandas • Jupyter Notebook

1. Project Overview

This beginner-level project explores a dataset of Netflix titles using Python and Pandas. The notebook loads the dataset, checks and handles missing values, compares the number of Movies and TV Shows associated with the United States, and examines movie maturity ratings.

2. Project Objectives

• Load the Netflix titles dataset into a Pandas DataFrame.

• Inspect the dataset and identify missing values.

• Remove rows containing missing values using dropna().

• Filter titles whose country field contains 'United States'.

• Count Movies and TV Shows in the filtered data.

• Explore the distribution of movie maturity-rating categories.

3. Dataset Information

Dataset file: netflix_titles.csv
The commonly used Netflix titles dataset contains 8,807 records and 12 columns: show_id, type, title, director, cast, country, date_added, release_year, rating, duration, listed_in, and description.

The country field can contain multiple countries for a title. This analysis searches for 'United States' anywhere in that field.

4. Tools and Technologies

Tool / Library	Purpose
Python	Programming language
Pandas	Data loading, cleaning, filtering, and counting
Jupyter Notebook	Writing and running analysis code
CSV	Dataset file format

5. Analysis Steps

Step 1 — Import libraries

```
import numpy as np
import pandas as pd
```

Step 2 — Load the dataset

```
netflix = pd.read_csv("netflix_titles.csv")
```

Step 3 — Inspect missing values

```
netflix.isnull().sum()
```

Step 4 — Remove rows containing null values

```
df_clean = netflix.dropna()
```

In the standard 8,807-row dataset, dropna() leaves 5,332 rows because it removes every row with at least one missing value. This is a strict cleaning choice.

Step 5 — Filter USA titles and count content types

```
us_content = df_clean.query(
    "country.str.contains('United States')",
    engine='python'
)
us_content['type'].value_counts()
```

Use the actual output from your notebook when reporting the Movie and TV Show counts.

Step 6 — Explore movie ratings

```
movies = df_clean.query("type == 'Movie'")
movies['rating'].value_counts()
movies['rating'].mode()
```

The rating column contains categorical maturity labels such as TV-MA, TV-14, PG-13, R, and PG. These are not numerical scores, so mean() is not appropriate. Use value_counts() to see the distribution and mode() to find the most frequent category.

6. Key Findings and Interpretation

• Missing values occur in several dataset columns.
• dropna() removes any row with one or more missing fields; in the standard dataset, the cleaned result contains 5,332 rows.
• The USA comparison counts titles in the dataset, not actual viewer activity.
• Movie ratings are maturity classifications, not numerical audience scores.

7. Project Folder Structure

```
Netflix-Data-Analysis/
■■■ Netflix_Assignment.ipynb
■■■ netflix_titles.csv
■■■ README.md (or README PDF)
```

8. How to Run the Project

• Install Python and Jupyter Notebook, or open the notebook in Google Colab.

• Keep the CSV file in the same working folder as the notebook, or update the file path.

• Open Netflix_Assignment.ipynb.

• Run the notebook cells from top to bottom.

• Review the displayed outputs for the cleaning and analysis steps.

9. Conclusion

This project demonstrates basic data analysis with Pandas: loading a CSV file, checking and cleaning missing data, filtering records, counting categories, and interpreting categorical values correctly. Use the notebook's actual outputs for exact counts in your submission.

Project: Netflix Data Analysis
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